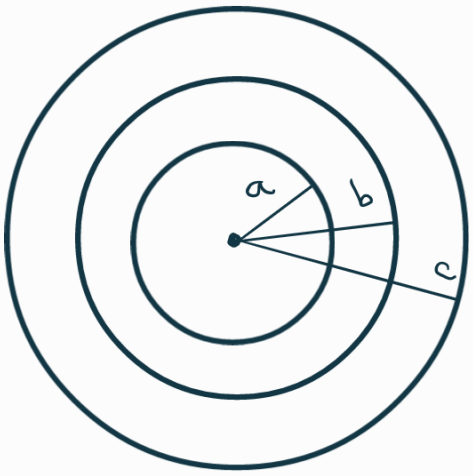
a) Calcular la magnitud del campo eléctrico para las regiones:

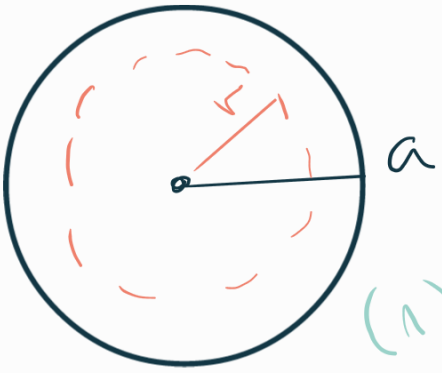


i) $r<a$
ii) $a<r<b$
iii) $b<r<c$
iv) $c<r$

Esfera maciza de radio 'a' con carga distribuida de forma homogénea $-3Q$, rodeada por un cascarón esférico de radio interno 'b' y externo 'c'. Carga $4Q$ cte.

i) $r<a$

$$\oint \vec{E}\cdot d\vec{A} = \frac{q_{enc}}{\varepsilon_0}$$



$$\int_0^{2\pi}\int_0^{\pi} E r^2 \operatorname{sen}\phi \, d\phi \, d\theta = \frac{q_{enc}}{\varepsilon_0}$$

(1)

$$E r^2 \int_0^{2\pi} d\theta \cdot \int_0^{\pi} \operatorname{sen}\phi \, d\phi = \frac{q_{enc}}{\varepsilon_0}$$

(1) $$Er^2(\theta)\Big|_0^{2\pi} \cdot (-\cos\phi)\Big|_0^{\pi} = \frac{q_{enc}}{\varepsilon_0}$$

$$Er^2 \cdot 2\pi \cdot 2 = \frac{q_{enc}}{\varepsilon_0}$$

$$Er^2 4\pi = \frac{q_{enc}}{\varepsilon_0}$$ (1)

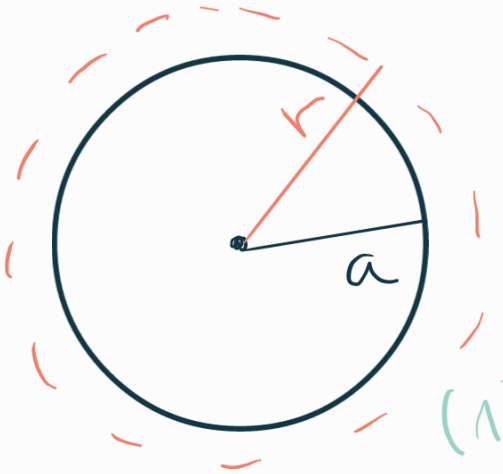
Para q_{enc}:

$$\frac{-3Q'}{-3Q} = \frac{\frac{4}{3}\pi r^3}{\frac{4}{3}\pi a^3} \Rightarrow -3Q' = -\frac{3Qr^3}{a^3}$$

$$Er^2 4\pi = \frac{-3Qr^3}{a^3 \varepsilon_0}$$

$$E = \frac{-3Qr}{4\pi \varepsilon_0 a^3} \quad [N/C]$$ (2)

ii) Para $a<r<b$



$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\varepsilon_0}$$

(1)

Sabemos que:

$$\oint \vec{E} \cdot d\vec{A} = \int_0^{2\pi}\int_0^{\pi} Er^2 \operatorname{sen}\phi \, d\phi \, d\theta = Er^2 4\pi$$

(1)

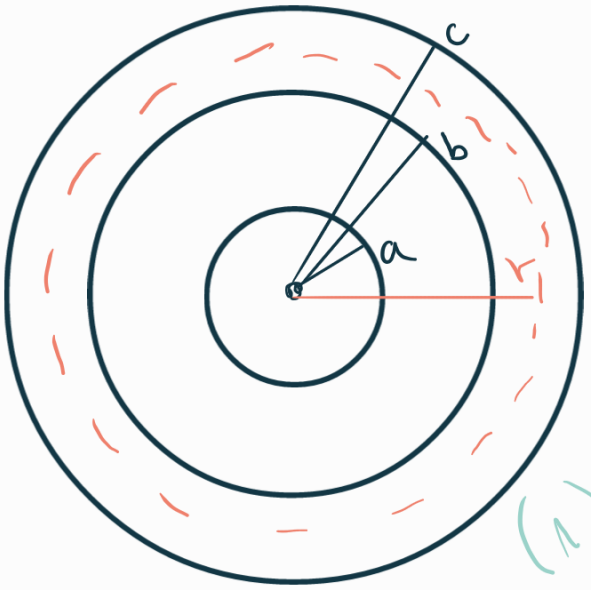
y $q_{enc} = -3Q$

Por lo tanto:

$$E r^2 4\pi = \frac{-3Q}{\varepsilon_0}$$

$$E = \frac{-3Q}{4\pi r^2 \varepsilon_0} \left[N/C\right] \quad (2)$$

iii) Para $b<r<c$



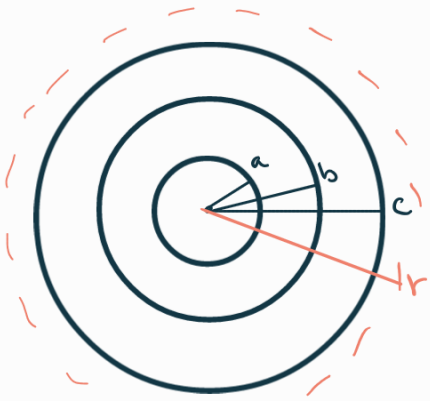
(1)

En esta región

$E = 0 \left[N/C\right]$ (2)

Porque el radio está dentro de un conductor.

iv) Para $r<c$



(1)

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\varepsilon_0}$$

Sabemos que

$$\oint \vec{E} \cdot d\vec{A} = \int_0^{2\pi}\int_0^{\pi} E r^2 \operatorname{sen}\phi \, d\phi \, d\theta = E 4\pi r^2 \quad (1)$$

y $q_{enc} = -3Q + 4Q = Q$

(2)

Por lo tanto

$$E\,4\pi r^2 = \frac{Q}{\varepsilon_0} \Rightarrow E = \frac{Q}{4\pi r^2\,\varepsilon_0}\left[\frac{N}{C}\right]$$

b) Si $a = 10\,cm$, $b = 12\,cm$ y $c = 15\,cm$.

Calcular la magnitud del campo eléctrico para $r = 11\,cm$ con $\varepsilon_0 = 8{,}85 \cdot 10^{-12}\left[C^2/N \cdot m^2\right]$

Es el caso de ii) $E = \dfrac{-3Q}{4\pi r^2\,\varepsilon_0}\left[N/C\right]$

(1)

Reemplazamos:

$$E = \frac{-3 \cdot 120 \cdot 10^{-6}\,C}{4\pi\,(0{,}11\,m)^2 \cdot 8{,}85 \cdot 10^{-12}\left[\frac{C^2}{Nm^2}\right]}$$

$$E = -2{,}675 \cdot 10^{8}\left[N/C\right]$$

(2)